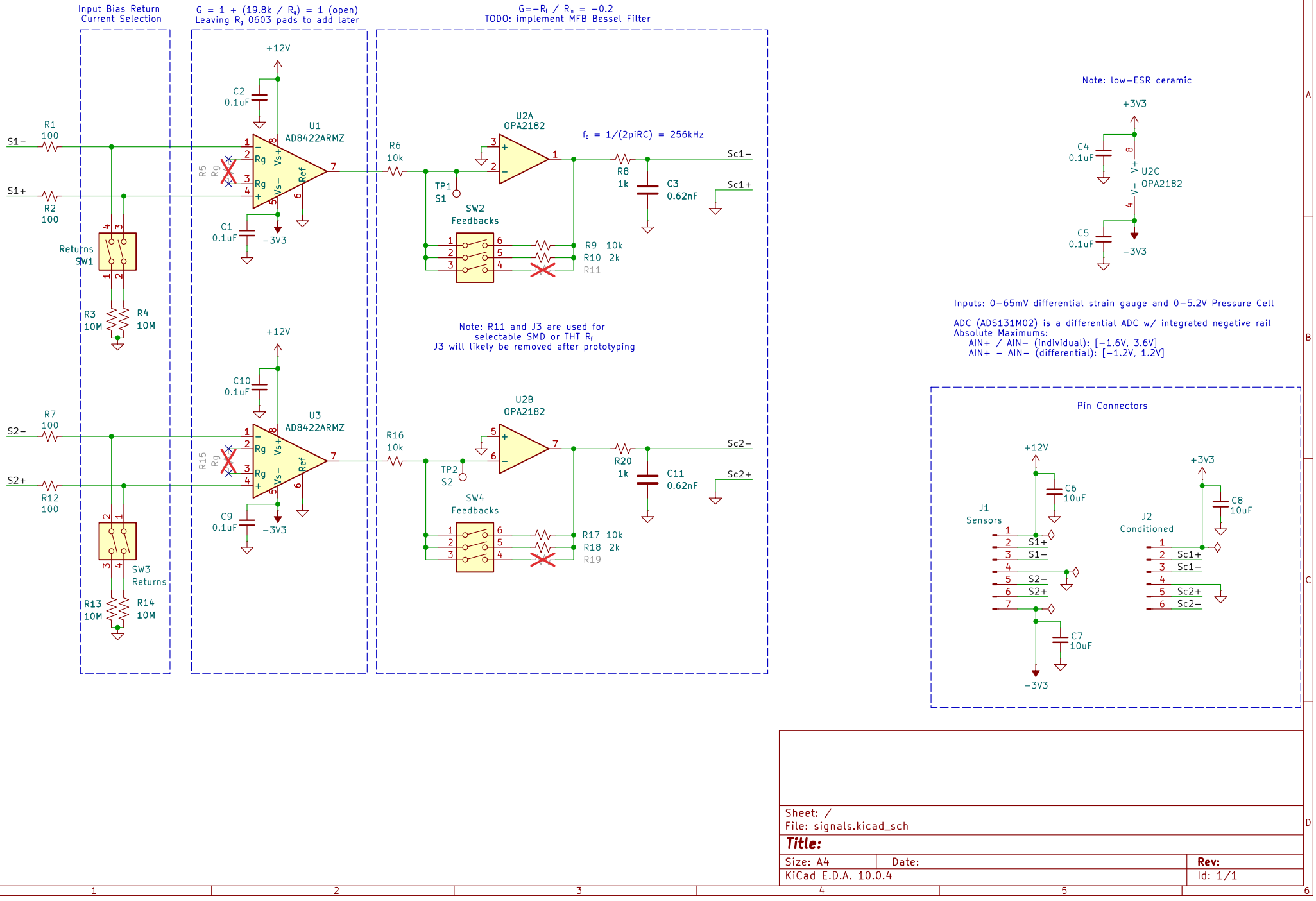

Input Bias Return Current Selection
G = 1 + (19.8k / Rg) = 1 (open)
Leaving Rg 0603 pads to add later
G=−Rf / Rin = −0.2
TODO: implement MFB Bessel Filter
S1−
S1+
S2−
S2+
R1 100
R2 100
R7 100
R12 100
Returns SW1
SW3 Returns
R3 10M
R4 10M
R13 10M
R14 10M
+12V
−3V3
C2 0.1uF
C1 0.1uF
C10 0.1uF
C9 0.1uF
U1 AD8422ARMZ
U3 AD8422ARMZ
R5 Rg
R15 Rg
R6 10k
R16 10k
U2A OPA2182
U2B OPA2182
fc = 1/(2piRC) = 256kHz
R8 1k
R20 1k
C3 0.62nF
C11 0.62nF
Sc1−
Sc1+
Sc2−
Sc2+
TP1 S1
TP2 S2
SW2 Feedbacks
SW4 Feedbacks
R9 10k
R10 2k
R11
R17 10k
R18 2k
R19
Note: R11 and J3 are used for selectable SMD or THT Rf
J3 will likely be removed after prototyping
Note: low−ESR ceramic
+3V3
C4 0.1uF
C5 0.1uF
U2C OPA2182
Inputs: 0−65mV differential strain gauge and 0−5.2V Pressure Cell
ADC (ADS131M02) is a differential ADC w/ integrated negative rail
Absolute Maximums:
AIN+ / AIN− (individual): [−1.6V, 3.6V]
AIN+ − AIN− (differential): [−1.2V, 1.2V]
Pin Connectors
J1 Sensors
J2 Conditioned
C6 10uF
C7 10uF
C8 10uF
Sheet: /
File: signals.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1